

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

16

Industry 4.0

Section Id :	640653106148
Section Number :	12
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	16
Number of Questions to be attempted :	16
Section Marks :	20
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653245509
Question Shuffling Allowed :	No

Question Number : 207 Question Id : 6406531529887 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406535121459. ✓ YES

6406535121460. ✗ NO

Sub-Section Number :

2

Sub-Section Id :

640653245510

Question Shuffling Allowed :

Yes

Question Number : 208 Question Id : 6406531529888 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Match Col A with Col B for landmark outcome of industrial era :

Col A (Industrial Era)	Col B (Major Outcome)
1.0	Digitization
2.0	CNC
3.0	Steam Engine
4.0	Mass production

Options :

6406535121461. ✖ 1.0:CNC, 2.0:Mass production, 3.0: Digitization, 4.0:Steam Engine
6406535121462. ✖ 1.0:Steam Engine, 2.0:mass production, 3.0: digitization, 4.0:CNC
6406535121463. ✖ 1.0:Mass production, 2.0:Steam Engine, 3.0: digitization, 4.0:CNC
6406535121464. ✔ 1.0:Steam Engine, 2.0:mass production, 3.0: CNC, 4.0: Digitization

Sub-Section Number :

3

Sub-Section Id :

640653245511

Question Shuffling Allowed :

Yes

Question Number : 209 Question Id : 6406531529889 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Identify the Industry 4.0 version of transportation.

Options :

6406535121465. ✖ IC engine
6406535121466. ✖ Electric car
6406535121467. ✔ Hyperloop

Sub-Section Number :

4

Sub-Section Id :

640653245512

Question Shuffling Allowed :

Yes

Question Number : 210 Question Id : 6406531529890 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Select the layout(s) which do not require general supervision:

Options :

6406535121468. ✖ Product layout

6406535121469. ✓ Process layout

6406535121470. ✗ Group layout

6406535121471. ✗ Fixed position layout

Sub-Section Number : 5
Sub-Section Id : 640653245513
Question Shuffling Allowed : Yes

Question Number : 211 Question Id : 6406531529891 Question Type : MCQ
Correct Marks : 1

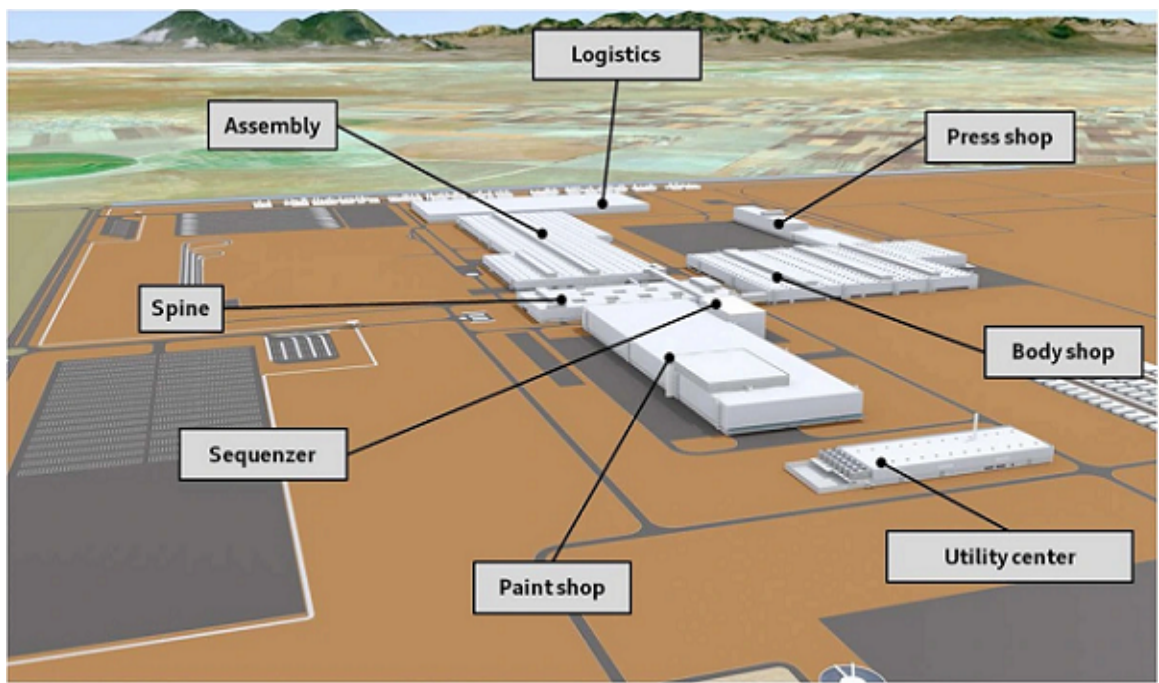
Question Label : Multiple Choice Question
Which of the following is an advantage of Process Layout?

- Options :**
- 6406535121472. ✗ Low material handling
 - 6406535121473. ✗ Total production time per unit is small
 - 6406535121474. ✗ Low training cost for the operators
 - 6406535121475. ✓ It can accommodate high variety of products

Sub-Section Number : 6
Sub-Section Id : 640653245514
Question Shuffling Allowed : Yes

Question Number : 212 Question Id : 6406531529892 Question Type : MCQ
Correct Marks : 1

Question Label : Multiple Choice Question
Following is the layout from an aircraft manufacturing company. Identify the layout from the figure below.



Options :

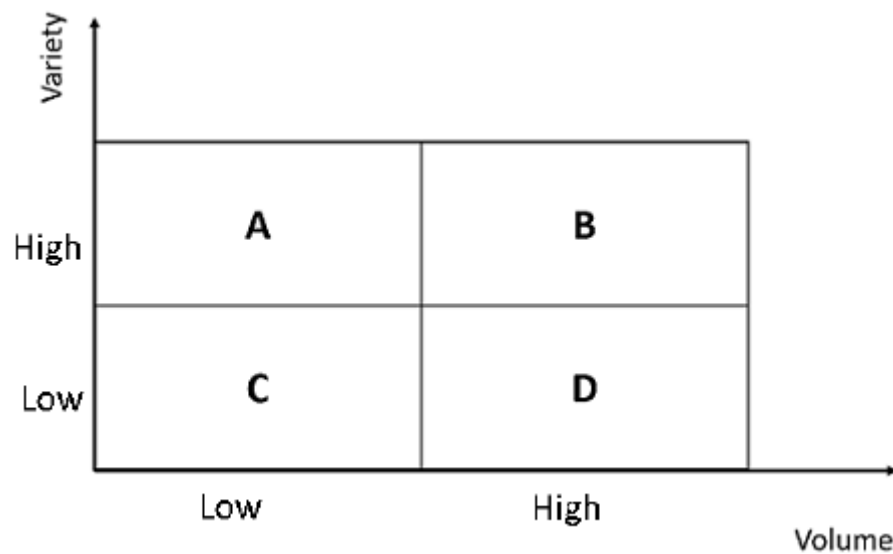
- 6406535121476. ✖ Product layout
- 6406535121477. ✖ Process Layout
- 6406535121478. ✖ Group layout
- 6406535121479. ✔ Fixed position layout.

Sub-Section Number : 7
Sub-Section Id : 640653245515
Question Shuffling Allowed : Yes

Question Number : 213 Question Id : 6406531529893 Question Type : MCQ
Correct Marks : 1

Question Label : Multiple Choice Question

Select the appropriate layout for Quadrants A, B, D in the following figure:



Options :

- 6406535121480. ✖ A: Group layout, B: Product layout , D: Process layout
- 6406535121481. ✖ A: Product layout, B: Process layout , D: Group layout
- 6406535121482. ✔ A:Process layout, B:Group layout , D: Product layout

Sub-Section Number : 8
Sub-Section Id : 640653245516
Question Shuffling Allowed : Yes

Question Number : 214 Question Id : 6406531529894 Question Type : MCQ
Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following is an advantage of Product Layout?

Options :

- 6406535121483. ✔ Small in-process inventory
- 6406535121484. ✖ It can accommodate high variety of products
- 6406535121485. ✖ High utilization of machines

6406535121486. ✖ Low investment

Sub-Section Number : 9
Sub-Section Id : 640653245517
Question Shuffling Allowed : Yes

Question Number : 215 Question Id : 6406531529895 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The Production rate of an assembly line is determined by its

Options :

6406535121487. ✔ Slowest station

6406535121488. ✖ Fastest station

6406535121489. ✖ Idle station

6406535121490. ✖ None of these

Sub-Section Number : 10
Sub-Section Id : 640653245518
Question Shuffling Allowed : Yes

Question Number : 216 Question Id : 6406531529896 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the correct statements from below:

Options :

6406535121491. ✖ Optimal solution in a facility location problem using Euclidean method will result in a region

6406535121492. ✔ Optimal solution in a facility location problem using Euclidean method rarely matches the center of gravity

6406535121493. ✔ Optimal solution in a facility location problem using Euclidean method will result in a point

6406535121494. ✖ Optimal solution in a facility location problem using Euclidean method matches the center of gravity in not of the cases

Sub-Section Number : 11
Sub-Section Id : 640653245519
Question Shuffling Allowed : Yes

Question Number : 217 Question Id : 6406531529897 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Identify the examples for Offshoring from the list below:

Options :

6406535121495. ✔ Micron Technology, a US based semiconductor giant, setting up a Supply

Chain Analytics team, in India.

6406535121496. ✔ Micron signs an agreement with Taiwan based TSMC Co., to manufacture silicon wafer for their ICs.

6406535121497. ✖ Western Australia's electricity network agency assigns Tata Consultancy Services at Pune, to support it with an overhaul of its legacy IT systems, aimed at delivering cheaper and more reliable power.

6406535121498. ✖ Tata Motors assigns the manufacturing of its braking system to TACO group.

Sub-Section Number : 12
Sub-Section Id : 640653245520
Question Shuffling Allowed : No

Question Id : 6406531529898 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (218 to 219)

Question Label : Comprehension

A factory works for two 8-hour shifts each day and produces a total of 30,000 electric bulbs using a set of workstations. There are 8 activities required to manufacture the bulb, such that the sum of all activity times is equal to 12 seconds. Answer the given subquestions:

Sub questions

Question Number : 218 Question Id : 6406531529899 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the cycle time (in seconds) at any workstations (round your answer to one decimal e.g. if your answer is 1.89 then enter your answer as 1.9)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.7 to 2

Question Number : 219 Question Id : 6406531529900 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the theoretical minimum number of workstations required?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

7

Sub-Section Number :

13

Sub-Section Id :

640653245521

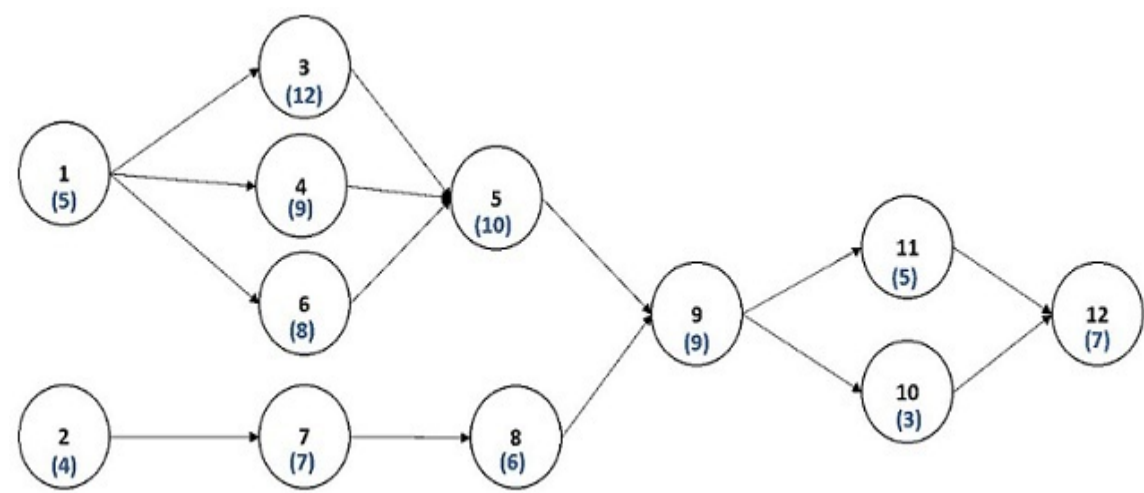
Question Shuffling Allowed :

No

Question Id : 6406531529901 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (220 to 224)

Question Label : Comprehension

Consider the following precedence diagram. It has 12 activities. The numbers in blue and withing the braces denote the respective activity times in minute.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 220 Question Id : 6406531529902 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the minimum cycle time of the system (answer in minutes without the unit e.g. if your answer is 10 min enter as 10)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

12

Question Number : 221 Question Id : 6406531529903 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the maximum cycle time of the system?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

85

Question Number : 222 Question Id : 6406531529904 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The factory works for one 8-hour shifts each day. Is it possible to achieve an output of 32 units/day

Options :

6406535121503. ✖ No

6406535121504. ✔ Yes

Question Number : 223 Question Id : 6406531529905 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the balance delay of the following layout (round it to two decimals)

(1,2) -> (3) -> (4) -> (5) -> (6,7) -> (8,9) -> (10,11,12)

(Instruction: if your answer is 14.52%, enter as 14.5)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

18.5 to 19.5

Question Number : 224 Question Id : 6406531529906 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the efficiency of the given layout?

(Instruction: Round off to two decimals without percentage sign. If your answer is 70.51%, enter as 70.5)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

80 to 81

Sub-Section Number : 14

Sub-Section Id : 640653245522

Question Shuffling Allowed : Yes

Question Number : 225 Question Id : 6406531529907 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

There are four demand points at A(1,2), B(2,3), C(3,5), D(4,1). The respective demand at these points are 7, 1, 3, 5. Select the optimal location based on cross-median approach.

Options :

6406535121507. ✓ (2,2)

6406535121508. ✓ (3,2)

6406535121509. ✓ (2.5,2)

6406535121510. ✗ (2,3)

Sub-Section Number : 15

Sub-Section Id : 640653245523

Question Shuffling Allowed : Yes

Question Number : 226 Question Id : 6406531529908 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Calculate the Euclidean distance between two machines located at the coordinates A(5,10) and B(2,6) in a shop floor.

[round off your answer to one decimal]

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.8 to 5.3

Sub-Section Number : 16

Sub-Section Id : 640653245524

Question Shuffling Allowed : Yes

Question Number : 227 Question Id : 6406531529909 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question